

The Reference Colour Space

So that accurate image colour relationships can be maintained as image data moves from capture (input) to an edited image ready for reproduction (output), image editing software uses an internal colour space as a reference for colour space translation calculations. This intermediate reference space is used because this minimises the number of device profiles and transforms required to translate from any input to any output.

Gradually over the years reference colour spaces and colour conversion transforms have moved towards better representations of perceptual linearity and towards greater accuracy. Perceptual linearity means that the individual coordinates that represent each of the measured colour shades the average human can discriminate are equi-distantly spaced within the colour space volume. As with all digital representations colour volumes are quantised, in this case into a number of points on a 3D grid. The spacing of the grid is determined by the bit depth used to represent it. For example if eight bit values are used for the height width and depth of a cube the cube will have 256 divisions on each axis and contain 16,777,216 equi-distant defined points. If the colour coordinates of all of the discernible shades coincided exactly with these grid points, 8 bit coordinates would be sufficiently accurate. However depending on the colour space used they may not be.

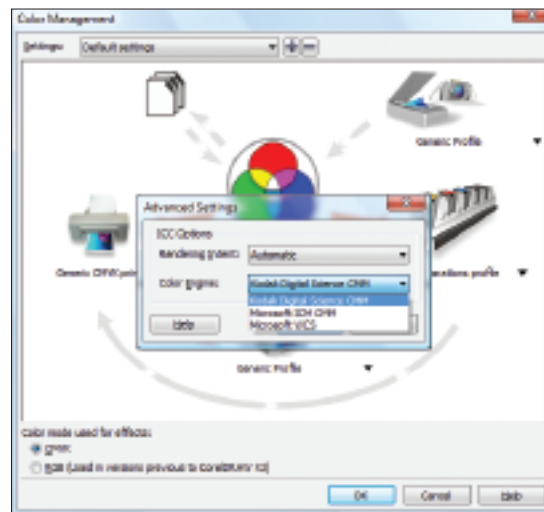
As described in a previous feature the earliest colour model in use is the CIE XYZ (1931) and this is based directly on measurements. The original measurements were made using a sample group of only seventeen people, although the results have since been confirmed using larger sample groups. Within the XYZ colour space some areas of colour discrimination of the human visual system are packed closely together and other areas are widely spaced. For example the greens occupy a large portion of the space while yellows occupy only a small portion. Based on the original XYZ measurements, CIE L*a*b* (1976) is one attempt to create a more perceptually linear colour space. Adobe products use CIE L*a*b* as their reference space while CorelDRAW Graphics Suite X4 uses sRGB.

The embedded colour management in Microsoft Windows from Windows 95 up to XP also uses the sRGB colour space for reference. This colour space was originally created by Microsoft and HP as a reference space for colour images on the world wide web. It is based on the typical device colour space for a CRT monitor. In this sense it is an idealised device dependant colour space and has quite a small gamut.

Colour Engines

The colour engine or Colour Management Module (CMM — Corel define this as Color Matching Module) is the piece of software that makes the calculations required to convert a colour image from one colour device and its associated colour space to another. These conversions are performed via the reference colour space. To make the conversion the colour engine uses the reference space, the source and destination colour space, the associated input and output device profiles and the users choice of rendering intent. Because of the complexity of the calculations involved and a variety of methods for exactly how the transformations are performed, there are a number of different colour engines in use.

Some image editing software provides a choice of which colour engine or CMM to use. For example by left clicking on the central colour icon in the colour management control panel in CorelDRAW (X3 or X4) the



The CorelDRAW Color Management control panel with the Advanced Settings for Rendering Intent and Color Engine selected. Three choices of Colour Engine are shown

user has the choice of either the Kodak Digital Science or Microsoft ICM 2.0 CMMs. With Microsoft Vista there is also the additional choice of — Microsoft WCS. In Adobe Photoshop, opening the Color Settings control panel from the Edit menu ([Shift] + [Ctrl] + [K]) with the More Options button selected, under Conversion Options — Engine, allows the user to choose either Adobe (ACE)

or Microsoft ICM. Differences in the algorithms used in different colour engines, coupled with the use of different reference colour spaces can result in visible differences in edited images when the same source file is processed by different image editing software.

Moving Colour Images Between Adobe And Corel

There some limitations on file format support between the competing products. The native file formats are: PSD for Photoshop, CDR for CorelDRAW and PSP for Paint Shop Pro. Paint Shop Pro X2 can save files in PSD format and CorelDRAW can save in AI Adobe Illustrator format. Adobe

Photoshop formats are more restricted, although it can save in one of the standard image formats such as JPG or TIF which Corel programs can open or import.

CorelDRAW cannot open JPG files, these must be imported using the File > Import menu choices. For an image to have the same colour appearance in CorelDRAW that it has in Photoshop the colour work flows must be set up with the same colour spaces and profiles.

Microsoft And Colour Management

Starting with Windows 95, in versions of Windows preceding Vista, Microsoft introduced the Image Color Management (ICM) and later ICM2, colour models, which are ICC based.

With Windows Vista, Microsoft, in cooperation with Canon, introduced a new colour model called the Windows Colour System (WCS). Microsoft say the ICC approach (i.e. the system used by just about every company except Microsoft, including Adobe and Corel) is fundamentally flawed, that problems with previous colour management systems are due to not using a perceptually accurate colour model, and that Vista with WCS provides this necessary perceptual accuracy (see Article References 3). Although by default WCS still uses sRGB (IEC 61966-2-1) as its reference colour space it does allow the use of other colour spaces, such as scRGB (IEC 61966-2-2). Colour accuracy is greater in scRGB because, although it uses identical color primaries and white/black points as the sRGB color space, it uses 16-bit floating point (half precision) linear values instead of the gamma compressed 8-bit integers used in sRGB. Microsoft call the colour engine used by WCS the Color Infrastructure and Translation Engine (CITE). It uses improved transform algorithms based on the CIECAM02 perceptual model to perform the colour calculations (see Article References 4).

Setting The Colour Controls

In Adobe Photoshop the the colour management controls are accessed from Edit > Color Settings (or [Shift] + [Ctrl] + [K]).

Adobe Photoshop and Corel Photo Shop have a setting for Working Space. This sets the default colour space that any un-tagged image files will be converted into when opened and also will be saved as. There are separate settings for RGB and CMYK files. Depending on the Policies settings, opening image files that are tagged with a colour space that is not the same as the working space can trigger an options warning.

Below the CMYK setting are choices for Dot Gain. Dot gain relates to the absorbency of the paper used for printing. Dots represented as a certain size in the image file tend to grow in size



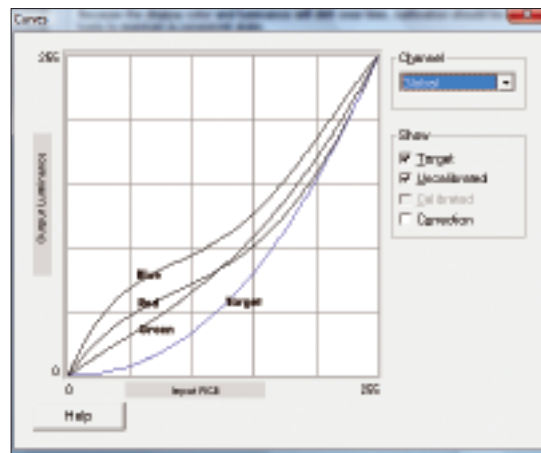
By default the CorelDRAW Color Management control panel shows all profile choices set to 'generic' with only the profiles converting from the internal space to the two printer types and from the desktop printer to the display, turned on

flow. In CorelDRAW X3, in X4 and in the associated Photo-Paint, the menu selection Tools > Color Management, launches a control panel displaying a graphical representation of the colour work flow (pressing the Alt o, Alt c key combinations also opens this window). The internal colour space is shown at the centre with five icons representing scanner or camera input, printing press output, display screen, desktop colour print output and file input/output surrounding it in an equally spaced ring. Drop down menus below each icon allow selection of profiles, while arrows running between the internal colour space and the device icons allow selection of different colour paths. Clicking on the arrows toggles the appropriate profiles on and off. Further arrows running between the two print icons and the display icon allow settings for soft proofing. Left clicking on all of the icons except the scanner/camera opens an Advanced settings control panel associated with that icon.

Display Calibration And Profiling

To be able to maintain any control over colour it is absolutely essential that the computer display is calibrated and correctly profiled. The display is the users visual window and reference on what's happening to colour at various stages in the colour editing process.

Calibration involves setting the display to known levels of brightness, contrast and colour balance. This is achieved through



This graph shows the un-calibrated R, G and B response curves for a laptop LCD display. Not only are these a long way from the desired target, but they also don't track each other. The desired target curve is represented by the blue line